VR. A Formal System.

An Axiom-Free Reconstruction of Arithmetic from Operations Alone

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Abstract

We present a formal system VR generated by **two** primitive *operations* $\{\varnothing,\ t\}$: the nullary base $\varnothing$ and the unary succession t. VR posits *no axioms of its own*: membership $\in$ is *defined* by recursion on t, the succession principle A3 is a *theorem*, and induction A4 is the *recursor* of the inductive term structure. (Versions through v1.1.0 listed a third primitive, a propositional implication $\to$ on a two-valued type; v1.1.1 removes it — it generated only a finite truth-function algebra used by no VR theorem, which is arithmetic on a 2-element type, not logic in any inferential sense. The logic VR reasons *with* — $\forall$, $\to$, $\leftrightarrow$, induction, Leibnizian equality — is metatheoretic, borrowed from the ambient framework, not generated by VR.) The von Neumann natural numbers are the unfolding of the succession operator t from $\varnothing$, the nullary base operation; membership $\in$ is a *defined* relation, not a primitive; equality is Leibnizian, and coincides with structural term-identity by a theorem. VR is arithmetically equivalent to first-order Peano arithmetic (PA), under the schematic reading of the Leibnizian quantifier, whence consistency relative to ZF follows as a metatheoretic corollary. "Axiom-free" is meant relative to the type-theoretic framework: VR adds no postulate beyond the free term-algebra (the nullary base and unary succession) and its recursor — this is exactly the axiom-free [] profile of the Lean 4 companion (Zenodo DOI 10.5281/zenodo.20324240), which formalises Part I.

Keywords: operational reconstruction of arithmetic, axiom-free system, foundations of mathematics, Leibnizian equality, von Neumann ordinals, Peano arithmetic, consistency.

Version note (1.1.1). A correction of scope: the third "primitive", the propositional implication $\to$ on a two-valued type $\{\mathsf{F}, \top\}$, is **removed**. It was load-bearing for no VR theorem (the Lean companion confirms: nothing in the arithmetic — membership, the von Neumann numbers, $+, \times, {}^\wedge$, the PA-equivalence — references it), and it is not "logic" in any inferential sense: a truth-table on a 2-element type is a finite function algebra, i.e. arithmetic. VR therefore neither needs nor generates a logic of its own; the logic it reasons with is metatheoretic (the ambient framework's). The arithmetic content — VR $\simeq$ first-order PA — is *unchanged*; this version removes a non-load-bearing appendage and the over-claim that VR founds its own propositional logic. Earlier version notes follow.

Version note (1.1.0). This is a minor-version revision of v1.0.3 (a change of thesis, not a patch: the title moves from "axiomatization" to "axiom-free reconstruction"). It (i) carries the operational reading ("there are only operations; $\varnothing$ is a nullary operation, not a posited thing") into the *body* of Part I — membership $\in$ is *defined* by recursion on t rather than presupposed, the gloss $t(x) = x \cup \{x\}$ is marked an abbreviation, and the metatheoretic status of the quantifiers is made explicit (the $\mathsf{F}/\varnothing$ "two registers and propositional-basis framing of that version is superseded by v1.1.1, which removes the logical register entirely); and (ii) **reframes the system as axiom-free** — the four named principles A1–A4 are reclassified as definitions and theorems over the free term-algebra, not postulates, matching the axiom-free Lean 4 companion. No statement is altered; only the status of A1–A4 is reclassified. The definition of $\in$ and the acyclicity Lemma are made explicit, the relative-consistency statement is marked metatheoretic, and the equality in the definition of $\in$ is identified as structural term-identity (related to Leibnizian equality by

`Eq_to_vrEq` and its immediate converse). Earlier editorial notes (v1.0.1–1.0.3) are retained in Part IV.

Contents

Part I. VR — the formal system

1. Primitives

VR is generated by **two** primitive *operations*:

- the **nullary** operation $\varnothing$ (the base — the act taking no operands; *not* a posited thing that "is empty");
- the **unary** operation t (succession).

Every object of VR is a *term* over this signature: $\varnothing$, $t(\varnothing)$, $t(t(\varnothing)), \ldots$. There are only operations and the terms they compose; no object is posited beyond what these operations generate. (In the Lean 4 companion this is literal: the objects form an inductive type `VRObj` with constructors `base` and `succ`.) This free term-algebra — together with its recursor (A4 below) — is the system's *only* foundational commitment; everything else, including the principles A3 and A4, is definition or theorem (§2).

Membership is defined, not primitive. The relation $\in$ used throughout is *not* a fourth primitive; it is defined by structural recursion on the generated term:

$$z \in \varnothing \;:=\; \bot, \qquad z \in t(y) \;:=\; (z \doteq y) \;\vee\; (z \in y). \tag{1}$$

Here $\doteq$ is **structural term-identity** ("is z the same generated term as y?"), which is prior to and independent of the Leibnizian equality of Def. 1: $\in$ is built on structural identity alone, so *no circularity* arises. (The $\bot$ and $\vee$ in (1) are the *metatheoretic* falsum and disjunction — `mem` is `Prop`-valued — not the removed two-valued type.) Structural identity implies Leibnizian equality (Lean: `Eq_to_vrEq`); the converse is immediate (instantiate the discriminating property $p := (x \doteq \cdot)$), so the two coincide — though only the direction actually used, `Eq_to_vrEq`, is recorded in the companion. Thus membership is decided by descending the term's t-structure and introduces no new primitive content. The familiar gloss $t(x) = x \cup \{x\}$ (used in A3) is likewise an *abbreviation*: $\cup$ and $\{\cdot\}$ are not primitives, and the substantive content of that gloss ($x \in t(x)$ and $x \subseteq t(x)$) is a *theorem* from (1) (§5). This keeps the "two operations" claim exact: $\in$, $\cup$, $\{\cdot\}$ are derived. (Lean: `mem` is (1); `A3_mem_self`, `A3_subset_succ` are theorems.)

2. The named principles A3, A4 (theorem and recursor)

Earlier versions stated four named principles A1–A4 as axioms. A1 and A2 governed the removed propositional layer; v1.1.1 drops them. What remains is purely operational: A3 (succession) is a *theorem*, A4 (induction) is the *recursor*. VR posits no axioms of its own.

A3 (Succession) — a *theorem*. t is the primitive succession operation, total on the generated objects. The content ascribed to A3 — written as the gloss $t(x) = x \cup \{x\}$ — is exactly: for every x, $x \in t(x)$ and $x \subseteq t(x)$. By the definition (1) of $\in$ both *hold as theorems* ($x \in t(x)$ since $x \doteq x$; $x \subseteq t(x)$ since $z \in x \Rightarrow z \in t(x)$). A3 postulates nothing beyond the totality of the constructor t, which is part of the term-algebra.

A4 (Induction) — the *recursor*. If P is a property of objects and (i) $P(O_0)$ and (ii) $\forall x : P(x) \to P(t(x))$, then $P(O_n)$ for all n; equivalently, the O_n exhaust the objects generated from $\varnothing$ via t. In the typed setting this is precisely the *recursor* (eliminator) of the inductive term-algebra — a derived principle, not a postulate (§13).

Axiom-free, precisely. The system's only foundational commitment is the free term-algebra: the objects are exactly the terms generated by the nullary base $\varnothing$ and the unary succession t, with the associated recursor. Given that framework, A3 and A4 add no postulational content: A3 is a theorem, A4 is the recursor. Hence VR is **axiom-free** — meaning it adds *no axioms beyond the type-theoretic framework* (which itself furnishes the inductive type and its recursor, and with them the potential infinity of $\mathbb{N}$). This is exactly the [] profile certified by the Lean companion: `#print axioms` reports no `propext`, `Classical.choice`, or `Quot.sound` for any object of Part I, and the inductive type and recursor are kernel primitives, not axioms. *What Peano postulates as axioms, VR derives.* The claim is not that VR assumes nothing — it assumes the term-algebra framework — but that it introduces no axioms of its own.

3. Logic is metatheoretic, not generated by VR

VR generates no logic of its own. The connectives and quantifiers used in its definitions and proofs — the $\forall p$ and $\leftrightarrow$ of Leibnizian equality (Def. 1), the $\exists$ of P2 (§9), the implication in the induction schema — belong to the ambient (first-order) metatheory and are read *schematically* (see §10 and the note to Def. 1).

What was removed. Versions through v1.1.0 carried a third primitive, a propositional implication $\to$ on a two-valued type $\{\mathsf{F}, \top\}$, with the claim that $\{\mathsf{F}, \to\}$ is functionally complete for classical propositional logic. That claim is true but inert: the two-element truth-function algebra is the *semantics* of classical propositional logic (the Boolean algebra $\mathbf{2}$), not the logic itself — it carries no consequence relation, no derivability, no soundness or completeness theorem — and no VR theorem uses it. Having the truth-*values* is not having the *logic*; and the logic VR actually reasons with is the metatheory's. The layer is therefore dropped in v1.1.1: at this level VR is pure arithmetic.

4. Definitions

Def. 1. **Leibnizian equality.** $x = y \;:=\; \forall p :\; p(x) \leftrightarrow p(y)$, p ranging over properties of objects. *Reading.* $\forall p$ is taken *schematically* — a schema over the (arithmetically definable) properties, matching the schema interpretation used for the PA-encoding in §10 — not as a genuine second-order quantifier. (A full second-order reading would make VR's equality second-order and the equivalence of §11 one with *second-order* PA, which is categorical and strictly stronger; the equivalence we prove, §11, is with *first-order* PA under the schematic reading. The $\leftrightarrow$ here is metatheoretic — the ambient logic, the only logic VR uses (§3). This $=$ relates to the structural identity $\doteq$ of (1) by `Eq_to_vrEq` and its immediate converse. Lean: `vrEq` over `VRObj→Prop`.)

Def. 2. **Distinctness.** $x \neq y := \neg(x = y)$.

Def. 3. $O_0 := \varnothing$. Def. 4. $O_{n+1} := t(O_n)$. Def. 5. The natural number n is identified with the ordinal O_n.

5. Acyclicity of $\in$, and $t(x) \neq x$

Lemma 1 (Antisymmetry and irreflexivity of $\in$)**.** *For all objects x, y: if $x \in y$ then $y \notin x$; in particular $x \notin x$.*

Proof. By induction on y (using (1)). Base $y = \varnothing$: $x \in \varnothing$ is $\perp$, vacuous. Step $y = t(w)$: suppose $x \in t(w)$ (i.e. $x \doteq w$ or $x \in w$) and, for contradiction, $t(w) \in x$. The lowering fact $t(a) \in b \Rightarrow a \in b$ (immediate induction on b; in the step $b = t(c)$ the case $t(a) \doteq c$ uses $a \in t(a)$, reflexivity in (1)) gives $w \in x$. If $x \doteq w$, then $w \in w$, contradicting the hypothesis at w; if $x \in w$, then with $w \in x$ the hypothesis at w forbids $w \in x$. Either way contradiction. Irreflexivity is $y := x$ against itself. (Lean: `mem_asymm`, `not_mem_self`, axiom-free.) ☐

Proposition 1. *$t(x) \neq x$ for every x.*

Proof. The property "contains x" holds at $t(x)$ (since $x \in t(x)$, A3) and fails at x (since $x \notin x$, Lemma 1); it distinguishes them, so $t(x) \neq x$ by Def. 2. ☐

6. The von Neumann numbers

$$O_0 = \varnothing, \quad O_1 = \{\varnothing\}, \quad O_2 = \{\varnothing, \{\varnothing\}\}, \quad O_3 = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, \ \ldots$$

Each O_n contains all preceding $O_0, \ldots, O_{n-1}$ (from (1) by induction; Lean: `O_mem_lt`).

7. Arithmetic

Addition $a + O_0 := a$, $a + t(b) := t(a + b)$; multiplication $a \times O_0 := O_0$, $a \times t(b) := (a \times b) + a$; exponentiation $a^{O_0} := O_1$, $a^{t(b)} := a^b \times a$. **Theorems** (by induction, A4): T1 commutativity, T2 associativity, T3 distributivity, T4 $O_1 + O_1 = O_2$. (On T1's structural cost and on T1–T4 not being needed for §11, see §16.)

8. Structure of the system

Layer	Content
Foundational commitment	the free term-algebra on $\{\varnothing$ (nullary), t (unary)$\}$ + recursor
Primitive operations	$\varnothing$, t; $\in$ defined by (1); $\cup, \{\cdot\}$ abbreviations
Named principles	A3 (thm), A4 (recursor) — *no postulates*
Logic	metatheoretic, borrowed from the ambient framework; NOT generated by VR (§3)
Equality	Leibnizian, schematic (Def. 1); $= \equiv \doteq$ by theorem
Objects & arithmetic	von Neumann O_n; $+, \times, ^\wedge$ (A4)

Part II. Equivalence of VR and Peano arithmetic

9. Direction VR → PA

The five Peano axioms (P1 $0 \in \mathbb{N}$; P2 $\forall n\, \exists S(n)$; P3 $S(n) \neq 0$; P4 $S(n) = S(m) \Rightarrow n = m$; P5 induction) hold in VR under $\mathbb{N} \mapsto \{O_n\}$, $0 \mapsto O_0$, $S \mapsto t$.

- **P1, P2.** $O_0 := \varnothing$ is a term; t is total on the generated objects. (§14: in the typed setting these are facts of the syntax, not theorems.)

- **P3.** "contains an element" fails at $\varnothing$ and holds at $t(O_n)$; by Def. 2 they differ.

- **P4** ($t(x) = t(y) \Rightarrow x = y$). From $t(x) = t(y)$ and (1) the members coincide, giving ($x \doteq y \vee x \in y$) and ($y \doteq x \vee y \in x$). By Lemma 1: the case $x \in y \wedge y \in x$ is excluded by antisymmetry; the mixed cases $x \doteq y \wedge y \in x$ and $x \in y \wedge y \doteq x$ substitute to $x \in x$, excluded by irreflexivity. Only $x \doteq y$ survives. (Lean: `P4_succ_inj` via constructor injectivity.)

- **P5** is A4.

10. Direction PA → VR

Encode VR objects as naturals by arithmetization (Gödel numbering): $\ulcorner\varnothing\urcorner := 0$, $\ulcorner t(x)\urcorner := \ulcorner x\urcorner + 1$ (one suitable choice), so $\ulcorner O_n\urcorner = n$ for this numbering. The defined $\in$ and the gloss-operations are primitive recursive on codes (Mendelson; Boolos & Jeffrey). A3 is a primitive recursive relation; A4 is PA induction. Leibnizian equality is read as the *schema* over arithmetical formulas — the schematic reading of Def. 1 — which makes the target *first-order* PA.

11. Equivalence theorem

Theorem 1. *Under the schematic reading of Def. 1, VR and first-order PA are arithmetically equivalent: the $\mathbb{N}$-theoretic content of one corresponds bijectively to that of the other.*

The Lean companion realises this as a concrete structural isomorphism $O : \mathbb{N} \to$ `VRObj` with inverse, preserving $0, \mathrm{succ}, +, \times, ^\wedge$ (`Theorem_11_VR_PA`); §15.

Which equality (for the foundations reader). Two notions coincide here and must not be conflated. The *schematic* Def. 1 — a first-order schema over arithmetically definable predicates — is what yields equivalence with *first-order* PA. The Lean companion's `vrEq` $:= \forall p : \mathrm{VRObj} \to \mathrm{Prop},\ p(x) \leftrightarrow p(y)$ is instead the *full* (impredicative) Leibniz equality; on the concrete carrier `VRObj` it coincides with the kernel's structural `Eq` (`Eq_to_vrEq` and its immediate converse). Hence `Theorem_11_VR_PA` is realised as an isomorphism $\mathbb{N} \simeq \mathrm{VRObj}$ over structural `Eq`, and the [] profile is honest (impredicativity of `Prop` is part of the type theory, not an axiom). The two equalities agree on this carrier but are not the same notion in general; the first-order PA-equivalence is the schematic reading.

12. Consistency relative to ZF (metatheoretic)

Corollary 1. *VR is consistent relative to ZF.*

Proof. PA is consistent relative to ZF (the von Neumann $\mathbb{N} \subseteq \mathrm{ZF}$ realises the PA axioms — classical). By Theorem 1, a contradiction in VR would yield one in PA, hence in ZF; if ZF is consistent, so is VR. □

Status. A *metatheoretic* corollary via the equivalence; *not* part of the Lean formalisation. The formalisation verifies Part I's theorems by the kernel and is axiom-free in Lean's sense; the relative-consistency statement is classical metatheory cited here. Chain: VR $\leftrightarrow$ PA $\hookrightarrow$ ZF.

Part III. Open questions

(1) **Number systems.** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ over VR. *Status:* VR-Numbers (DOI 10.5281/zenodo.20272743), as operational superstructures over VR's naturals without posited ordered pairs.

(2) **Type theory.** VR as an initial t-algebra; $t =$ `succ`, $\varnothing$ =zero, A4 = the `Nat` eliminator; Leibnizian vs. path equality (HoTT).

(3) **Tarski / semantic closure.** Whether a level hierarchy via iteration of t evades undefinability (a direction, not a result).

(4) **Proof assistant.** *Status:* completed in Lean 4 (DOI 10.5281/zenodo.20324240); Part IV.

Part IV. Notes from the Lean 4 formalisation

Part I is fully formalised in Lean 4 (DOI 10.5281/zenodo.20324240): every theorem is verified by the kernel, no mathlib arithmetic is used, and every object is axiom-free in Lean's sense. The notes mark where the formal layer is more revealing; several are now folded into the body (v1.1.0) and kept here for the record.

13. A4 is the recursor, not an axiom. VR-objects form an inductive type (`base`, `succ`); A4 is its recursor — derivable, not postulated.

14. P1 and P2 are absorbed by typing. O_0 is a well-typed term; succ is total. Both are facts of the syntax. Only P3–P5 retain content as theorems.

15. The Equivalence Theorem (§11) has a constructively stronger form. The metatheoretic equivalence is realised as an explicit type isomorphism with inverse, preserving all operations.

16. T1–T4 are not needed for the equivalence. The isomorphism $O(m+n) = \texttt{vadd}(Om)(On)$ and its analogues close by induction on the right argument, using only the recursive symmetry of `Nat`/`vadd` etc. (T1 is structurally costlier than T2–T3: addition recurses in its second argument, so $O_0 + b = b$ and $t(a) + b = t(a + b)$ are separate lemmas — a cost the symmetric notation hides.)

17–18. The propositional layer is removed (v1.1.1). Versions through v1.1.0 carried a two-valued type `VRBool` with a truth-function `impl` (the former $\to$) and a biconditional `viff`. The Lean companion confirms what motivated the removal: these were referenced by no theorem of the arithmetic (membership, the von Neumann numbers, $+, \times, ^\wedge$, the PA-equivalence), and `#print axioms` on the entire arithmetic chain is unchanged ([]) without them. The only $\leftrightarrow$ that survives is the metatheoretic `Iff` of Leibnizian equality (Def. 1). VR generates no logic of its own; the logic it reasons with is the ambient framework's.

19. $\in$ is defined; equality is structural; acyclicity is structural. $\in$ is the recursive definition (1) (`mem`); the $=$ in it is the kernel's structural `Eq`; structural identity implies Leibnizian equality (`Eq_to_vrEq`), the converse being immediate, so they coincide (only `Eq_to_vrEq` is recorded, the converse not being needed); the set-content of A3 is a theorem (`A3_mem_self`, `A3_subset_succ`); acyclicity (Lemma 1) is by structural induction with no measure (`mem_asymm`, `not_mem_self`).

20. The system is axiom-free. Collecting §13 (A4 = recursor) with A3 (theorem from (1)): VR has *no proper axioms.* `#print axioms` returns [] for every object of Part I (no `propext`, `Classical.choice`, or `Quot.sound`). The sole foundational commitment is the inductive term-algebra and its recursor — kernel primitives, not VR axioms. This is the precise sense of "axiom-free": VR adds nothing to the type-theoretic framework; what PA postulates, VR derives.

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